

Advanced Full Stack Assignment: AI-Driven Software Engineering

This assignment evaluates your ability to leverage Generative AI as a "Force Multiplier." We aren't just looking for someone who can write code; we are looking for an engineer who can build the **foundation layer** that enables AI to generate a production-grade application that should be maintained by various teams.

The Objective

You must develop a comprehensive "AI Blueprint" (consisting of system design definitions for AI driven development, and an initial prompt) that, when executed, generates a fully functional, high-quality eCommerce platform.

Core Technology Stack

- **Frontend:** React.js (with beautiful, modern UI/UX)
- **Backend:** Node.js (Express)
- **Database:** MySQL
- **Design:** Any modern styling library is allowed (Tailwind CSS, Framer Motion, etc.), but it must be visually "premium."

Required eCommerce Features

- **Authentication:** Full Login/Sign-up flow with JWT or Session management.
- **Product Catalog:** Search, filter, and detailed product views.
- **Cart & Checkout:** Persistent cart and a multi-step checkout process.
- **Account Section:** Order history and profile management.

The AI Orchestration Requirements

Your task is to submit the "Engine" you built to create this site. You must define:

1. **Engineering Guidelines & Constraints**
Define the conventions the AI should consistently follow across the project. This includes decisions such as language preferences (e.g., TypeScript), architectural patterns, folder structure, naming conventions, error handling strategies, and coding standards.
2. **Capability Definitions**
Describe the functional building blocks and domains the AI can leverage while generating the application. This may include reusable modules, integration patterns, authentication strategies, data access layers, and UI composition approaches.
3. **initial.md (Bootstrap Prompt)**
The primary instruction set that initiates the project generation. This should establish the

architecture, define the stack, and guide the AI through building the core features of the system end-to-end.

Submission Structure

Your Git repository must be organized as follows:

Component	Requirement
1. Resulted Code	The final, working codebase in its completed state.
2. AI Boilerplate	The initial.md file and all AI guideline files used to generate the project.
3. README.md	Documentation of manual interventions . Specify exactly what you had to fix by hand and why the AI failed to handle it.
4. AI interactions Documentation	Documentation of all the Prompts, queries (also search queries in google, chatgpt questions ...), planning and architecture inquiries you used for this projects. Also document the Models you used in which step and why you used those model. Write all the plugins and tools you used

Evaluation Criteria

- **Prompt Engineering Rigor:** How "zero-touch" is your generation process?
- **Architectural Integrity:** Does the AI-generated code follow best practices?
- **Critical Thinking:** Your ability to identify the "AI-Gap" (where human intervention is more efficient than prompt tuning).
- **Design Aesthetic:** The visual quality of the resulting React frontend.

Show us how you lead the machine, rather than just how you follow it. Good luck!